

A 1-V 450-nW Fully Integrated Programmable Biomedical Sensor Interface Chip

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Abstract—This paper presents a fully integrated programmable biomedical sensor interface chip dedicated to the processing of various types of biomedical signals. The chip, optimized for high power efficiency, contains a low noise amplifier, a tunable bandpass filter, a programmable gain stage, and a successive approximation register analog-to-digital converter. A novel balanced tunable pseudo-resistor is proposed to achieve low signal distortion and high dynamic range under low voltage operations. A 53 nW, 30 kHz relaxation oscillator is included on-chip for low power consumption and full integration. The design was fabricated in a 0.35 μm standard CMOS process and tested at 1 V supply. The analog front-end has measured frequency response from 4.5 mHz to 292 Hz, programmable gains from 45.6 dB to 60 dB, input referred noise of $2.5 \mu\text{V}_{\text{rms}}$ in the amplifier bandwidth, a noise efficiency factor (NEF) of 3.26, and a low distortion of less than 0.6% with full voltage swing at the ADC input. The system consumes 445 nA in the 31 Hz narrowband mode for heart rate detection and 895 nA in the 292 Hz wideband mode for ECG recording.

Index Terms—Biomedical electronics, low noise amplifier, low-power circuit design, sensor interface, tunable pseudo-resistor.

I. INTRODUCTION

ADVANCES in CMOS technologies, communications, and low power circuit design techniques have spurred considerable interests in wearable biomedical devices, a phenomenon which can potentially revolutionize the healthcare industry. One of the building blocks in a wearable device is the sensor interface, whose design has to meet a set of stringent requirements for both analog and digital blocks. The main challenges in the design of the analog front-end circuits are associated with the nature of physiological signals. The amplitudes of these signals are in the order of tens of μV to tens of mV and the frequencies span from DC to a few kHz, as illustrated in Fig. 1 [1]. In order to process these signals, the analog front-end should be designed with low input referred noise, reconfigurable bandwidth, and programmable gain to accommodate the weak signals and the high dynamic range. Furthermore, the electrode-skin interface usually introduces a high DC component in the input signal which has to be removed by a high-pass filter with

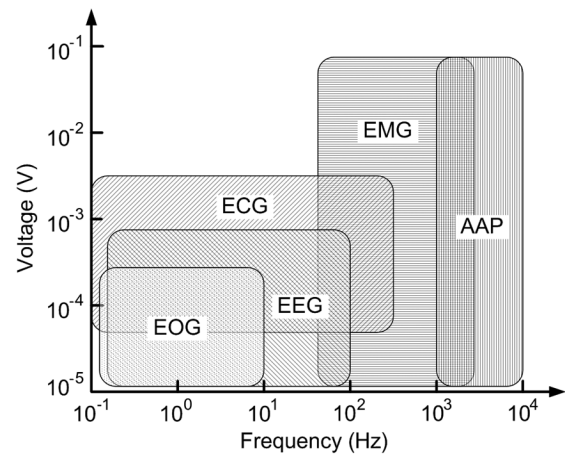


Fig. 1. Voltage and frequency ranges of some common physiological signals, where EOG, EEG, ECG, EMG, and AAP refer to the electrooculography, the electroencephalography, the electrocardiogram, the electromyography, and the axon action potential, respectively.

cut-off frequency below 1 Hz. While it is easy to implement such a high-pass filter with external capacitors and resistors [2], it is more economical to integrate the filter on-chip. Switched-capacitor and MOS-bipolar pseudo-resistor [3] are two possible candidates which achieve good trade-offs between performance, power, and area as demonstrated in [4], [5]. It is interesting to note that pseudo-resistor based approach not only reduces the area but also facilitates the design of reconfigurable band-pass filters [6]–[9]. The only drawback in the existing tunable pseudo-resistors is the large variation of resistances under positive and negative biasing conditions. This leads to a serious DC biasing offset at the output node, especially in low-voltage operation.

This paper, while addressing some of these issues, describes the design of a fully integrated programmable low-power sensor interface IC (first presented in [10]). The chip contains a low noise tunable bandwidth front-end amplifier (TB-FEA), a programmable gain amplifier (PGA) and a 12-bit successive approximation register analog-to-digital converter (SAR-ADC). A novel balanced tunable pseudo-resistor structure is introduced for the adjustment of high-pass corner frequency while the low-pass corner frequency is realized by the -3 dB roll-off of the operational transconductance amplifier (OTA) of the TB-FEA. The overall gain is programmed through the proposed flip-over-capacitor feedback in the PGA. The TB-FEA achieves a NEF of 3.26, which is among the lowest published values to date. In addition to an on-chip crystal oscillator, a 53 nW on-chip relaxation oscillator is implemented for further power reduction. This

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allows the chip to offer two operational modes, i.e., low power and regular modes. Under the low power mode for heart rate detection, the chip consumes 445 nW by setting the bandwidth to 31 Hz and the relaxation oscillator activated. The power dissipation increases to 895 nW in the regular power mode for ECG signal recording when the bandwidth is adjusted to 292 Hz and crystal oscillator is used.

The paper is organized as follows: Section II describes the system architecture of the low power sensor interface chip. The detailed circuit designs of the analog front-end and ADC are presented in Sections III and IV, respectively. Section V reports the measurement results, and Section VI concludes the paper.

II. SYSTEM DESIGN

In portable or implantable biomedical systems where power consumption is of primary concern, it is necessary to reduce the number of active building blocks and maximize the power efficiencies of each block. On the system level, this is accomplished by packing as many signal processing functions as possible into one active block while incorporating power management into the design. However, we find that this conventional system level approach does not necessarily lead to optimal power efficiency [11]. To illustrate this, we assume that the maximum frequency of input signal is $f_{\max}$, the S/H circuit works at Nyquist rate of $2f_{\max}$, and the bandwidth of front-end amplifier is f_{3dB} . We further assume that the effect of higher frequency poles/zeros in the amplifier is negligible, and the amplifier output goes through a first-order linear settling without slewing in each sampling period as illustrated in Fig. 2. To obtain a less than 1/2 LSB (least significant bit) tracking error for a rail-to-rail sine wave input $A \sin(2\pi f_{\max} t)$, f_{3dB} and $f_{\max}$ need to satisfy

$$\begin{aligned} \text{Tracking_Error} &= \left. \frac{dA \sin(2\pi f_{\max} t)}{dt} \right|_{\max} \cdot \frac{\eta}{2f_{\max}} \\ &\cdot e^{-\frac{(1-\eta)}{2f_{\max}} \cdot 2\pi f_{3dB}} < \frac{1}{2} \cdot \frac{2A}{2^n} \\ \Rightarrow \frac{f_{3dB}}{f_{\max}} &> \frac{\ln(2^n \pi \eta)}{\pi(1-\eta)} \end{aligned} \quad (1)$$

where n is the resolution of the quantization module, and η determines the ratio of holding period in each conversion cycle. Generally, a larger η means low ADC conversion rate, and often gives rise to more relaxed design complexity and reduced power overhead at the ADC. A moderate value of 0.5 for η in a 10-bit ADC would thereby suggest f_{3dB} to be at least 5 times as wide as $f_{\max}$. This requirement is hard to satisfy in the conventional single amplifier sensor interface design where the bandwidth of amplifier is set to $f_{\max}$ in order to perform low-pass filtering. To maintain an acceptable tracking error in the single amplifier design, the time left to ADC is shortened significantly, leading to unnecessarily high conversion speed and increased power consumption of ADC [5].

In view of this, we propose a new low power system architecture that differs from the conventional approach, as shown in Fig. 3. The analog front-end consists of two active blocks: a low noise preamplifier integrated with a reconfigurable band-pass filter, and a programmable gain stage that drives the sample

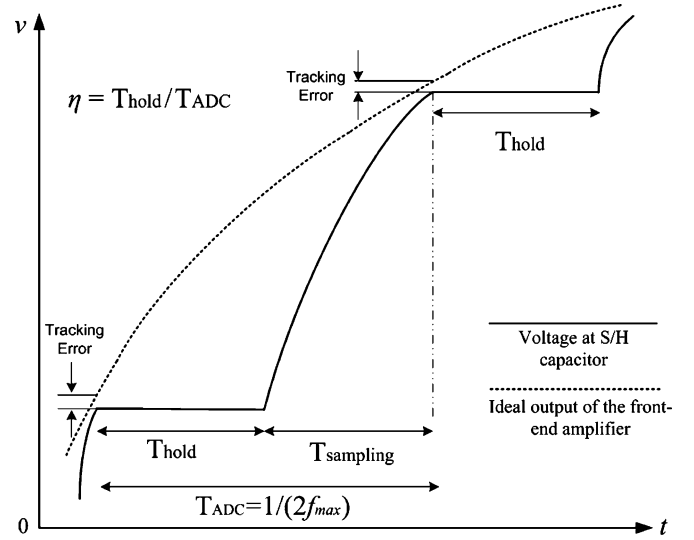


Fig. 2. Tracking error of the system.

and hold (S/H) circuit. A SAR-ADC is chosen as the quantization module due to its relatively good tradeoffs between power efficiency, conversion accuracy and design complexity. The clock of the entire chip is generated from either one of the two on-chip oscillators. The crystal oscillator guarantees superior clock accuracy, but draws more current and requires an external quartz crystal while the relaxation oscillator draws less current at the cost of large clock jitters. This programmable dual-clock scheme allows flexible system configuration in low power applications. The benefits of the proposed architecture are threefold. First, the linear tracking speed of the front-end output is no longer limited by the effective bandwidth of the input signal. This allows us to reduce the conversion speed of ADC. Second, the optimum power allocation between PGA and ADC is achievable by properly selecting biasing current of PGA and η based on (1). The third benefit is precise gain tuning. In the single amplifier approach, the bandwidth and gain tuning features are implemented within one closed-loop amplifier [6]. The fixed gain-bandwidth product suggests that bandwidth adjustment modifies the gain and vice versa. This issue does not exist in the proposed architecture.

Given the proposed system architecture, it is necessary to determine the gain, dynamic range, and power budget in order to achieve the best power efficiency. The overall gain is set by the amplitude of the physiological signal and the power supply voltage, whereas the desired dynamic range is determined by the accuracy requirement. For example, in a 1 V supply ECG monitoring system, the required gain varies from 200 to 1000 corresponding to possible ECG amplitudes of 1~5 mV. Generally, the gain of PGA should be at least one order of magnitude smaller than that of preamplifier to relax the noise and power requirements of PGA. Hence, a gain of 100 for the preamplifier would be suitable. In ECG applications, the desired dynamic range can be up to 60 dB, corresponding to μ V level of input-referred noise for the preamplifier and approximately 10-bit resolution for the ADC. Considering the resolution loss due to static/dynamic errors, the design target for the ADC is set to 12 bits.

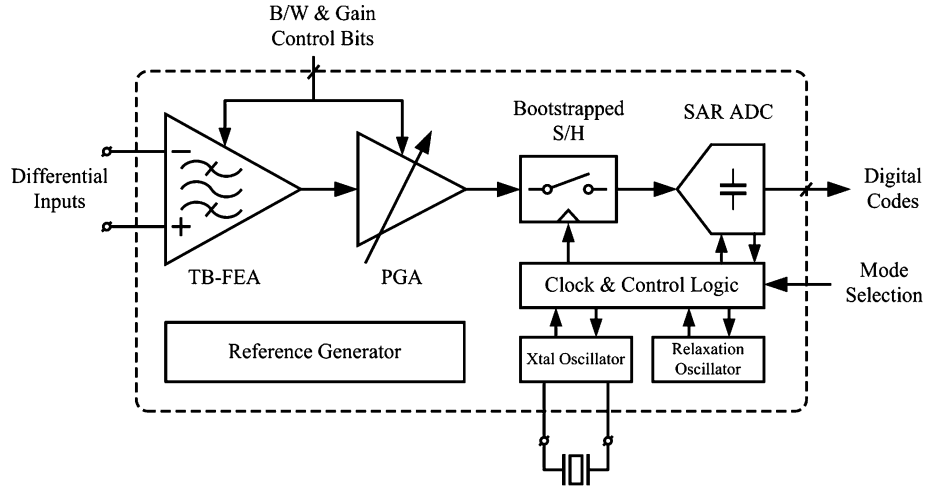


Fig. 3. Proposed low power system architecture.

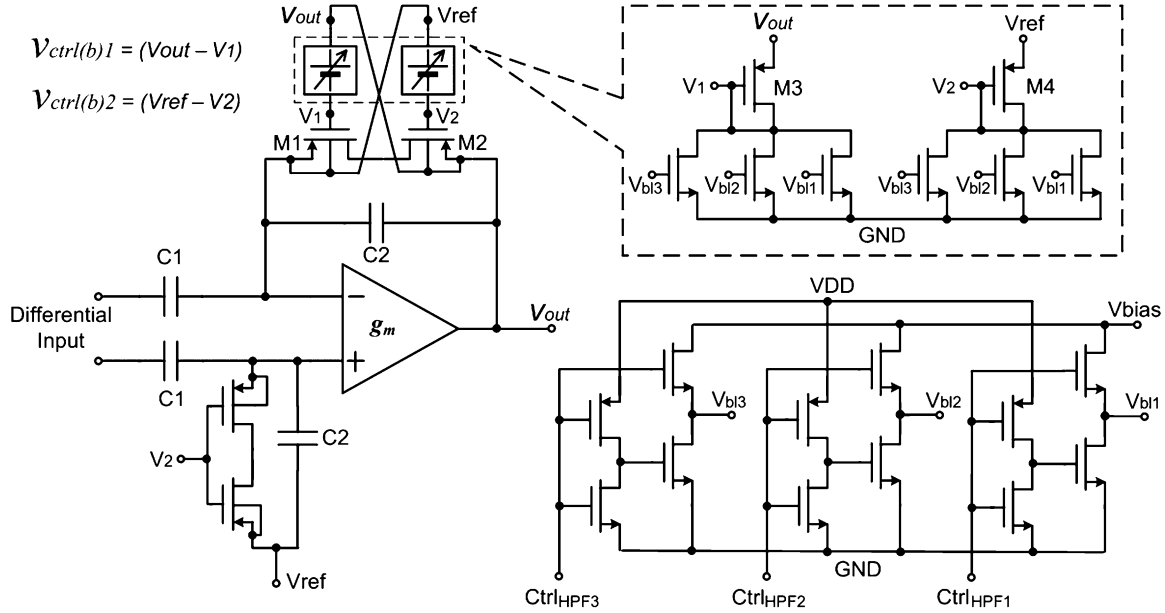


Fig. 4. Schematic of the TB-FEA with balanced tunable pseudo-resistors.

III. THE ANALOG FRONT-END

A. Tunable Bandwidth Front-End Amplifier

The proposed TB-FEA acts as a low noise amplifier as well as a reconfigurable bandpass filter. The schematic is shown in Fig. 4, where the low-pass corner frequency is adjusted by changing the bandwidth of the OTA, and the high-pass corner frequency is controlled by the balanced tunable pseudo-resistors. The midband gain of the TB-FEA is determined by the capacitance ratio C_1/C_2 . The DC levels of the OTA input and output are set in the middle of the supply rail to facilitate the maximum output swing.

1) *Balanced Tunable Pseudo-Resistor*: To effectively reject the large DC component, a high-pass filter is built into the front-end amplifier. To minimize the number of external components, tunable pseudo-resistors are usually employed to achieve a reconfigurable high-pass feature. However, the existing tunable pseudo-resistor [7]–[9], as shown in Fig. 5(a),

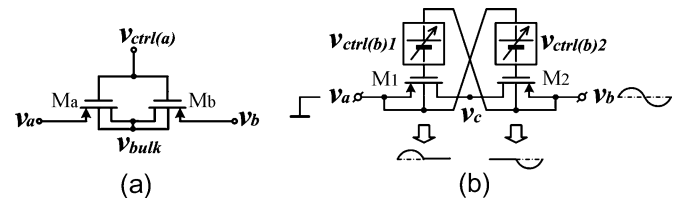


Fig. 5. (a) Schematic of the existing tunable pseudo-resistor. (b) Schematic of the proposed tunable pseudo-resistor.

exhibits asymmetric and nonlinear resistances when the voltage across the pseudo-resistor varies. This property is clearly demonstrated in Fig. 6 where the voltage across the pseudo-resistor, $(v_b - v_a)$, sweeps from -1 V to 1 V at different gate control voltages, $v_{ctrl(a)}$. Note that the simulations in Fig. 6 are carried out without considering parasitic imperfections of the transistors. The asymmetric resistances lead to unavoidable signal-dependent output shift as well as early clippings at

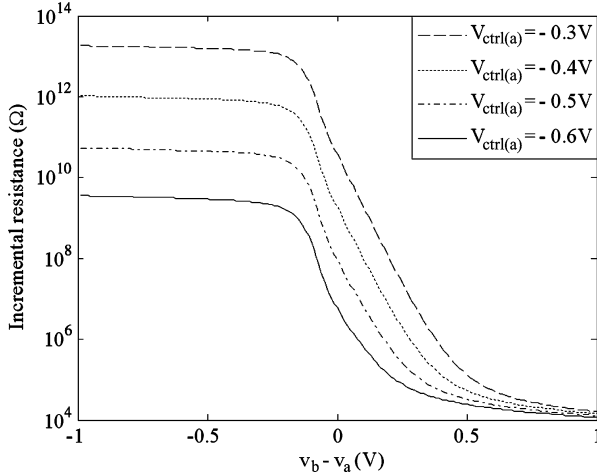


Fig. 6. Simulated resistance of the existing tunable pseudo-resistor biased at $v_a = 0$ V and v_b swept from -1 V to 1 V.

one of the power rail as the signal level increases. This is the primary factor which limits the dynamic range in low-voltage circuits. The nonlinearity of resistance, as perceived by signals sweeping over the slope of the curve in Fig. 6, introduces additional distortions. Furthermore, the junction leakage current from n-type well to the substrate in the order of sub pA or larger may considerably bias the high impedance node v_{bulk} and induce further unbalanced resistances of the pseudo-resistor.

To solve the above mentioned problems, we propose a fully balanced tunable pseudo-resistor structure, as shown in Fig. 5(b). The simulated resistances at different terminal voltages exhibit favorable symmetry, as shown in Fig. 7. The balanced property of the proposed tunable pseudo-resistor is explained by fixing v_a at a reference voltage and applying a sine wave on top of the reference voltage to v_b . During the positive half cycle of the sine wave, the gate of M_1 and node v_b are charged up, while the gate of M_2 and node v_a remain virtually constant. During this period, the voltage at v_b tends to turn on M_2 while keeping M_1 off, thereby the terminal voltage applies to M_1 only. Conversely, during the negative half cycle, the gate voltage of M_1 is pulled down below the reference level and tends to turn on M_1 , thereby the voltage swing is now left to the virtually shut M_2 . Consequently, each transistor works as an active resistor for only half of each cycle. The control voltages $v_{ctrl(b)1}$ and $v_{ctrl(b)2}$ in Fig. 5(b) are formed by $(v_{out} - v_1)$ and $(V_{ref} - v_2)$, respectively, as indicated in Fig. 4. Their values are defined by the gate to source voltages (v_{gs}) of M_3 and M_4 , respectively, which are in direct association with the drain currents of M_3 and M_4 (designed to be identical) and independent of v_{out} and V_{ref} . Since the M_1 and M_2 are turned on alternatively, $v_{ctrl(b)1}$ and $v_{ctrl(b)2}$ effectively set the v_{gs} for M_1 and M_2 . In this way, the resistance of pseudo-resistor is not affected by the change of v_{out} . To tune the resistance, 3-bit digital controlled current sources are used to adjust the current passing through M_3 and M_4 via $Ctrl_{HPF1} \sim Ctrl_{HPF3}$, which in turn determine the values of control voltages and set the high-pass corner frequency. Note that V_{bias} is a fixed voltage reference provided by the internal reference generator. It is worth noting that the maximum voltage swing across the

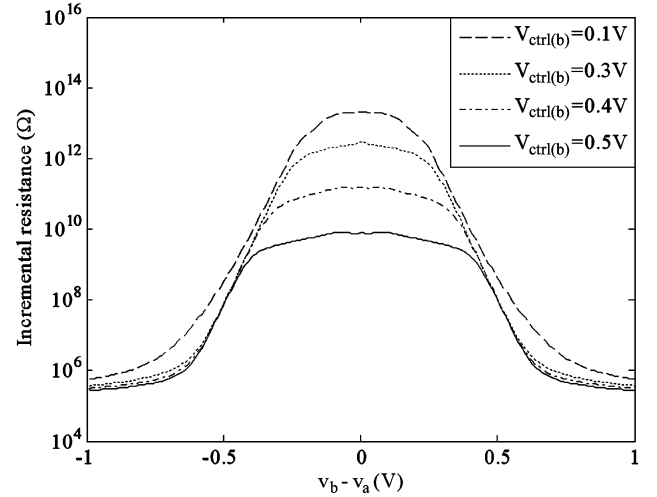


Fig. 7. Simulated resistance of the proposed tunable pseudo-resistor biased at $v_a = 0$ V and v_b swept from -1 V to 1 V. $v_{ctrl(b)}$ here equals $v_{ctrl(b)1}$ and $v_{ctrl(b)2}$ in Fig. 5(b).

tunable pseudo-resistor should be kept small in order to avoid large changes in resistance. In our design, the output voltage falls into ± 0.25 V, within which the generated resistances are almost constant and is sufficient to meet the gain requirement.

2) *Low Noise Operational Transconductance Amplifier (OTA)*: The low-pass corner frequency of the TB-FEA is determined by the -3 dB cut-off frequency of the OTA. There are two ways to change the bandwidth of the OTA: varying the load capacitance and adjusting the transconductance by means of changing the transistors bias current. However, as reported in [2], changing the capacitive load gives no DC current saving to narrow bandwidth applications. The better approach to program the low-pass corner frequency is to control the bias current of the OTA. Fig. 8 shows the low noise OTA employing the structure of a typical two-stage Miller opamp. The dominant pole of the OTA, which is the low-pass corner frequency of the TB-FEA, is adjusted by the 3-bit-controlled bias currents at both input and output stages. The OTA incorporates a push-pull output scheme by adopting M_{10} (M_{10A} , M_{10B} and M_{10C}) and C_2 (C_{2A} , C_{2B} and C_{2C}) [12], where the slew rate is not limited by the quiescent current of the OTA and the driving capability of the OTA is greatly enhanced. Transistors $M_{10A} \sim M_{10C}$ act as pseudo-resistors which operate in deep triode region and provide large resistance values in the order of 10^{12} ohms. The V_{bias} in Fig. 8 is the same V_{bias} as in Fig. 4.

It is well known that the input referred noise of the overall system relies on the noise performance of the front-end amplifier, thus minimizing the noise figure is one of the most important tasks in the design of the OTA. The input referred thermal noise of the OTA is given by

$$\overline{V_{n,total}^2} = \frac{16}{3} kT \frac{1}{g_{m1,2}} \left[1 + \frac{g_{m3,4}}{(1 + g_{m3,4} r_{o1}) g_{m1,2}} + \frac{g_{m5,6}}{g_{m1,2}} \right] + \frac{8}{3} kT \frac{1}{(g_{m8} + g_{m9})(g_{m1,2} R_{out1})^2}. \quad (2)$$

It is obvious that $g_{m1,2}$ plays the most important role in minimizing the input referred noise. In low power design, it may

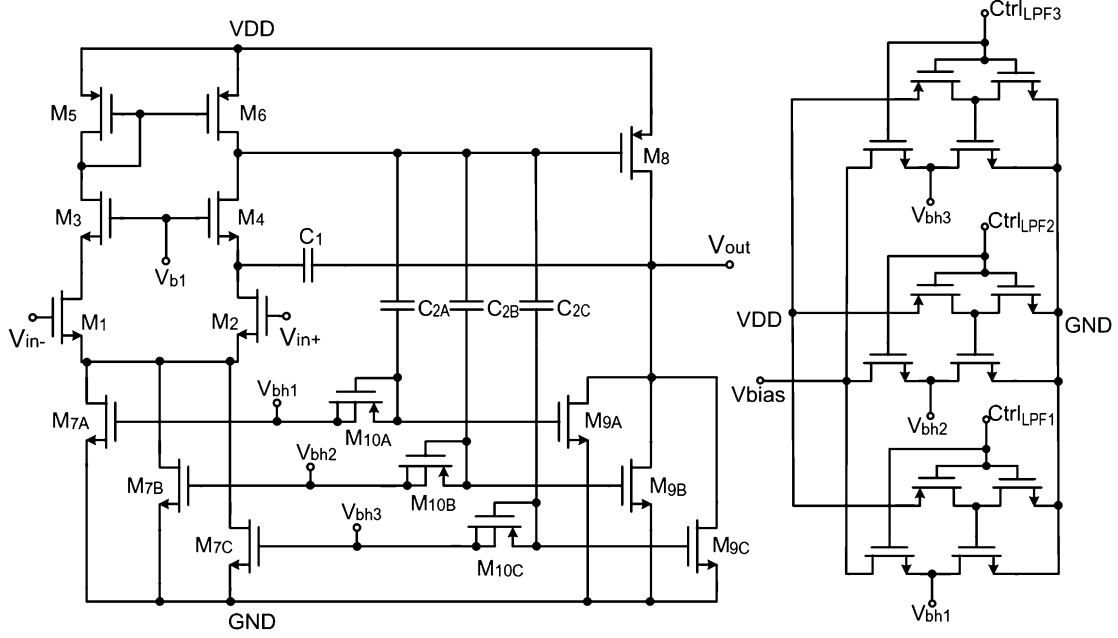


Fig. 8. Schematic of the low noise tunable bandwidth OTA.

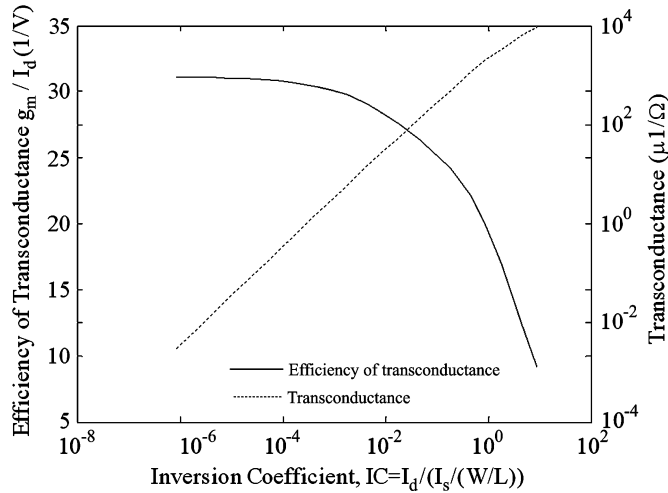


Fig. 9. Efficiency of transconductance versus inversion coefficient for input pair of the OTA.

not be efficient to boost transconductance by a large current. Maximizing the current efficiency becomes the key factor in the optimization of noise-to-power trade-off. In order to get a high transconductance while maintaining relatively high efficiency of transconductance, it would be better for the operation status of the input transistors to fall into the rolling off region of the transconductance efficiency curve shown in Fig. 9. We set the target power consumption for the overall system to be less than $1 \mu\text{W}$, so each input transistor gets maximum 150 nA bias current. Under these constraints, the input pair is designed with a high transconductance efficiency of about 30, where the transconductance is maximized within the power limit. A further improvement in the input referred noise is from the class AB output stage as the transconductance of the output stage increases from g_{m8} in the common source configuration to $(g_{m8} + g_{m9})$ in the class AB output scheme. To minimize the flicker

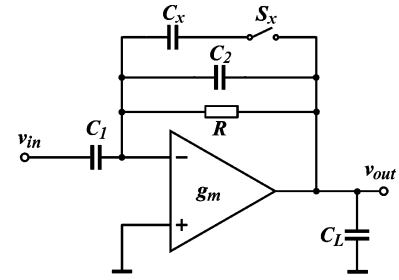


Fig. 10. Concept of the conventional PGA.

noise of the OTA, pMOS input pair is usually employed. However, due to the high threshold voltage of the pMOS transistor, an nMOS input pair is adopted here and the flicker noise is suppressed by choosing large gate area for the input transistors.

B. Programmable Gain Amplifier (PGA)

As mentioned in Section II, a PGA is introduced in the proposed architecture to regulate the gain and to drive the S/H circuit. The voltage gain of a close-loop amplifier can be adjusted by varying its feedback factor. A common approach is to use a switch to connect or disconnect a feedback path, as illustrated in Fig. 10. However, this scheme may not hold at very low frequency when the reactance of C_x becomes comparable to the off-state resistance R_x of the corresponding control switch. The close-loop gain can be estimated more precisely by

$$G = \frac{C_1}{C_2} \cdot \frac{sC_x R_x + 1}{sC_x R_x + 1 + C_x / C_2}. \quad (3)$$

Since C_x is comparable to C_2 , the resultant non-overlapping zero-pole pair distorts the frequency response with a small ripple. Given C_x in pF range and R_x approximately in 10^{13} ohm , the distortion occurs around sub-1 Hertz to a few Hertz, interfering with the high-pass function of the TB-FEA.

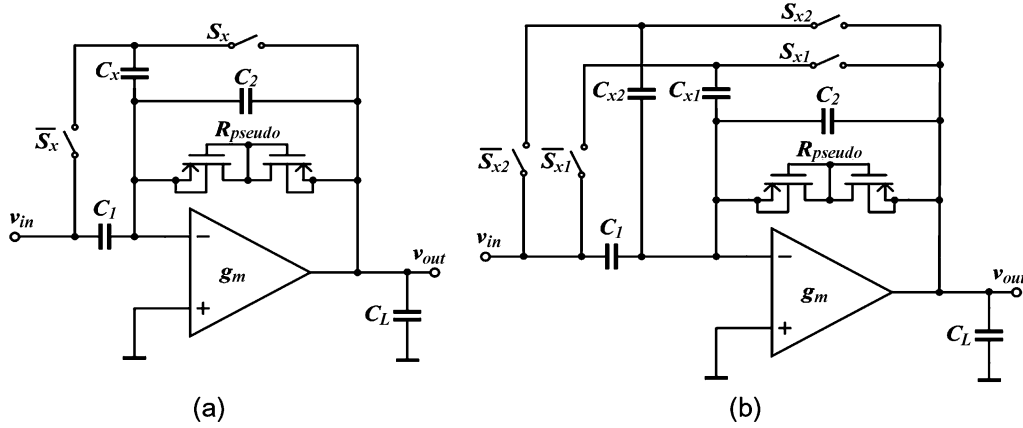


Fig. 11. (a) Concept of the proposed PGA with flip-over-capacitor. (b) Schematic of the PGA with two sets of C_x .

TABLE I
CONFIGURATION OF THE CONTROL SWITCHES OF PGA
FOR EACH GAIN SETTING

Gain of PGA	S_{x1}	$\bar{S}_{x1}$	S_{x2}	$\bar{S}_{x2}$
2	ON	OFF	ON	OFF
3	ON	OFF	OFF	ON
5	OFF	ON	ON	OFF
11	OFF	ON	OFF	ON

To correct the low frequency distortion, the “flip-over-capacitor” scheme is proposed as shown in Fig. 11(a). There are two control switches S_x and $\bar{S}_x$ associated with each flip-over-capacitor C_x , and the status of the two switches is always complementary. Hence, C_x functions either as a part of the input capacitor or as a part of the feedback capacitor according to the states of the two control switches. By doing so, the off-state resistance R_x of the corresponding switch is always excluded from the feedback loop and behaves only as a negligible load to the TB-FEA and the PGA, posing no threat to the frequency response of the system. Two sets of C_x are implemented in the PGA as shown in Fig. 11(b). By flipping over each C_x to either input or output node, four gain settings can be achieved, as shown in Table I. The capacitance values of C_1 , C_2 , C_{x1} , and C_{x2} are 8 pF, 1 pF, 2 pF, and 1 pF, respectively.

The DC biasing points of the PGA are regulated by a fully balanced pseudo-resistor with a fixed resistance. The resistance of the pseudo-resistor must be large enough to ensure that the resultant high-pass corner frequency is lower than the lowest high-pass corner frequency of the TB-FEA. To ensure high driving capability, the OTA implemented in the PGA adopts a similar topology as the one in the TB-FEA as it can provide large slew rate with low power consumption. The bias currents for the OTA are evenly distributed between the input and output stages to obtain wide bandwidth and safe phase margin.

IV. THE ADC BACKEND

A. Overall Architecture

Fig. 12(a) illustrates the architecture of the designed 12-bit SAR-ADC. The binary search array serves as the sampling capacitor during every tracking interval and as the DAC during every conversion interval. The DAC search result is successively

compared to a fixed reference voltage to determine each bit. This scheme eases the comparator design by relaxing its input range requirement and eliminating the dynamic offset problem. Depending on the real-time accuracy requirements, the mode selection bit selects either the crystal or the relaxation oscillator to drive the successive approximation (SA) timing sequence generator and in turn the overall ADC. The actual timing sequence is shown in Fig. 12(b), where η is set at 13/16 (≈ 0.8), providing the most optimal power partitioning for the overall system.

B. On-Chip Relaxation Oscillator

To further eliminate the power overhead produced by the crystal clock generator, a less accurate but more power efficient relaxation oscillator is designed. As illustrated in Fig. 13, the capacitor C is linearly charged up by a fixed current I , until V_C reaches the threshold level V_{REF} that is produced by a reference generator. A reset switch then shorts the capacitor to ground and triggers another charging cycle. Given the reset current is much higher than I , the cycle period T is dominated by the charging process, and is given by

$$T \approx \frac{V_{REF}}{I} \cdot C \quad (4)$$

In our design, the target oscillation frequency is 30 kHz. The current and capacitor are set to 20 nA and 500 fF, respectively. The generated pulse train is then regulated by a toggle flip-flop to produce a 50% duty ratio clock.

From (4), it can be seen that the clock generated from the relaxation oscillator is determined by two absolute parameters I and C , and thereby may deviate considerably from the design target due to process and ambient condition variations. Circuit-level compensation is possible but at the expense of complexity and increased power consumption. A more convenient and efficient approach is to perform system-level calibration at startup and periodically if needed (depending on ambient conditions), by temporarily switching on the crystal oscillator as a reference, and calculating the clock ratio for result correction.

V. MEASUREMENT RESULTS

The proposed design was fabricated in a 0.35 μm standard CMOS process with $V_{thn} + |V_{thp}| = 1.15$ V and the chip oc-

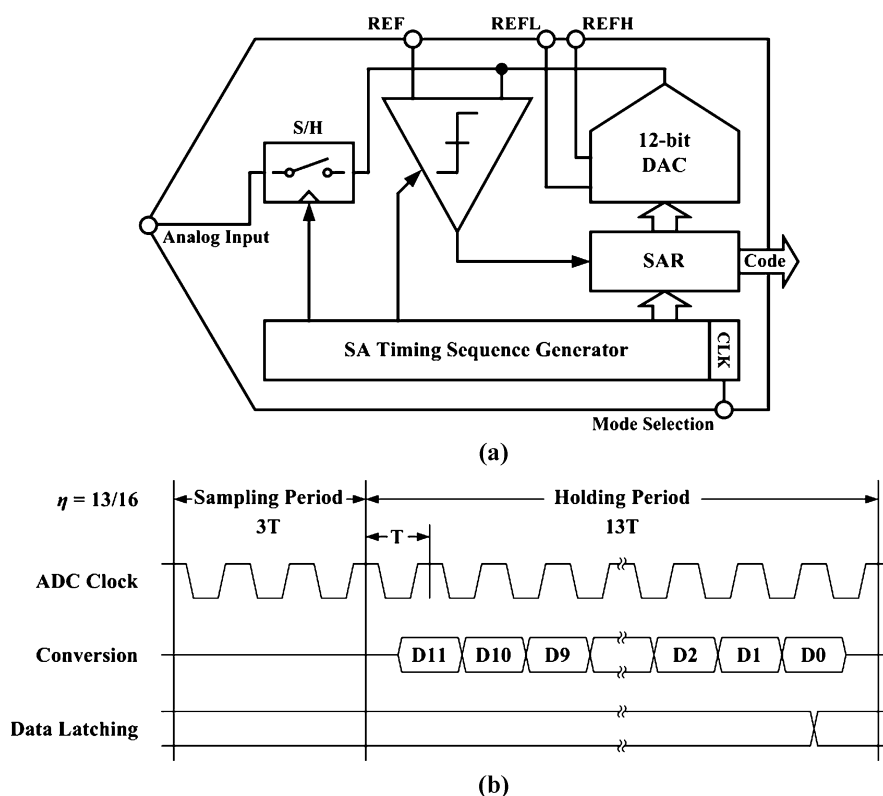


Fig. 12. (a) Architecture of the 12-bit SAR-ADC. (b) Timing sequence of the 12-bit SAR-ADC.

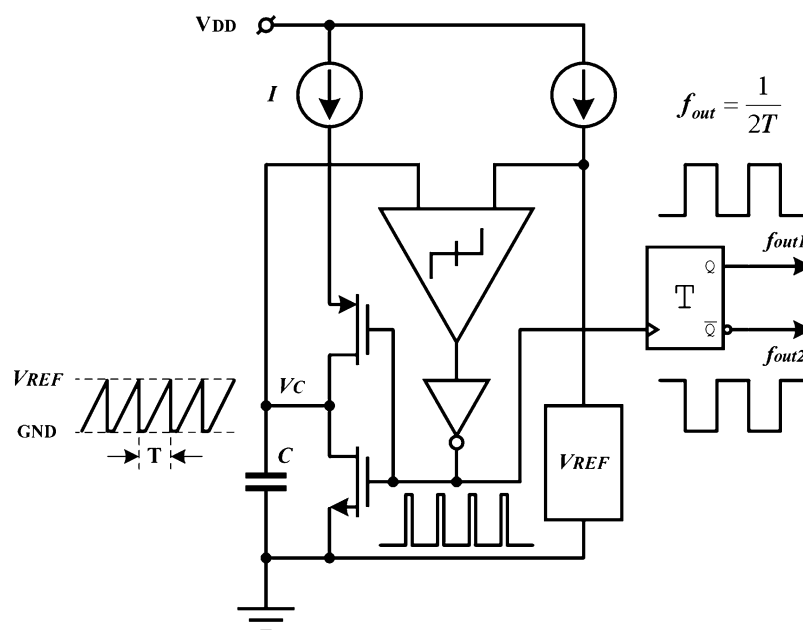


Fig. 13. Simplified schematic of the on-chip relaxation oscillator.

cupies 1 mm² silicon area excluding pads, as shown in Fig. 14. The system is designed and tested under a 1 V supply and the measurement results are summarized in the following sections.

A. Analog Front-End

The measured frequency response of the TB-FEA is shown in Fig. 15. It realizes a midband gain of 39.2 dB with tunable bandwidth control, where the lower corner frequency can be

adjusted from 4.5 mHz to 3.6 Hz and the upper corner frequency can be adjusted from 31 Hz to 292 Hz. The voltage gain of the system can be adjusted by changing the gain of the PGA, where 4 level voltage gains of 45.6 dB, 49 dB, 53.5 dB, and 60 dB can be selected, as plotted in Fig. 16. Since the current control is employed to tune the low-pass cut-off frequencies, the current dissipation of the TB-FEA reaches its minimum of 33 nA when the narrowest bandwidth of 31 Hz is selected and maximum of

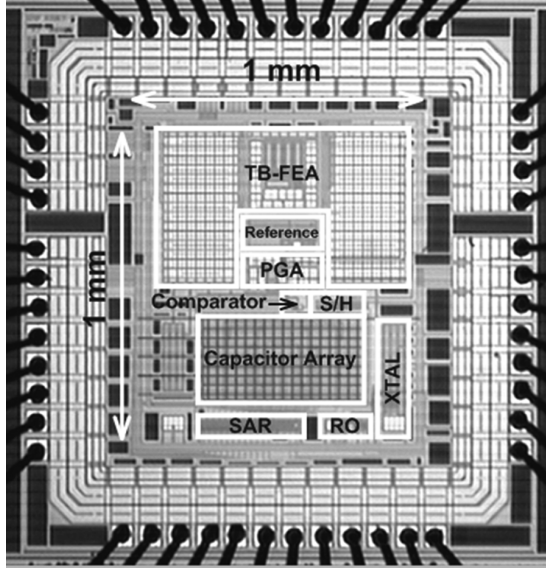


Fig. 14. Chip microphotograph.

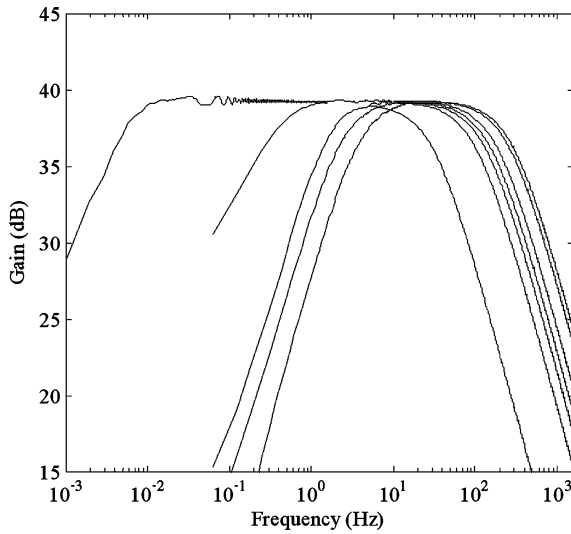


Fig. 15. Frequency response of the TB-FEA.

337 nA when the widest bandwidth of 292 Hz is chosen. The input referred noise of the TB-FEA, measured using the Agilent dynamic signal analyzer 35670A, with the maximum bandwidth is shown in Fig. 17. The large gate area of the input transistors results in a low corner frequency of flicker noise. The rms value of the input referred noise is calculated by integrating the curve in Fig. 17 across the equivalent noise bandwidth $f_{n,eq}$ given by

$$f_{n,eq} = \frac{\pi}{2} f_{-3 \text{ dB, TB-FEA}} \approx 460 \text{ Hz.} \quad (5)$$

The rms value of the input referred noise is $2.5 \mu\text{V}$ integrated from 0.05 Hz to 460 Hz. To measure the noise to power trade-off of the amplifier, the parameter NEF introduced in [13] is adopted. The NEF is an evaluation of an amplifier noise performance taking its quiescent current and bandwidth into consideration. Lower NEF means better noise to power trade-off. The state-of-the-art designs have demonstrated NEF

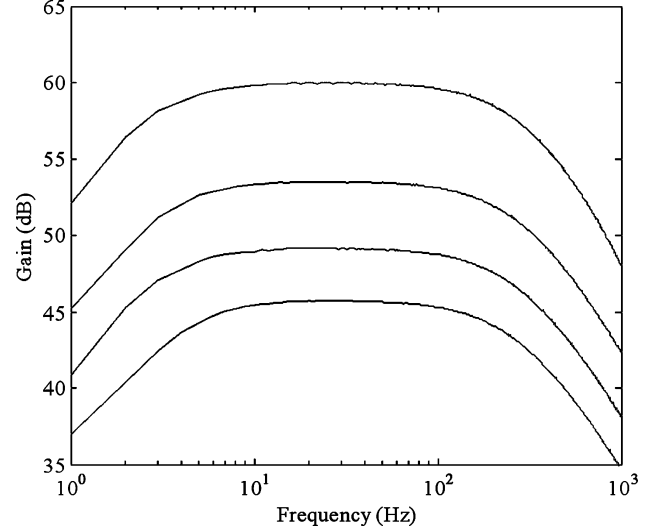


Fig. 16. System gain adjustment with consistent bandwidth (one bandwidth setting is chosen for illustration).

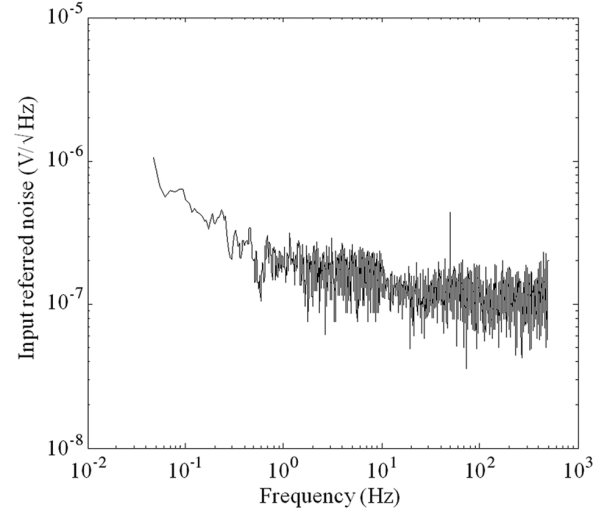


Fig. 17. Input referred noise of the TB-FEA.

in the range of 3.8 to 9.2 [2]–[6]. Due to the high transconductance efficiency and large gate area of the differential pair transistors, the NEF of this design reaches 3.26, which is among these lowest values published to date. The measured total harmonic distortion (THD) against the output amplitude of the analog front-end is plotted in Fig. 18. Since most of the signal degradation occurs at the tunable pseudo-resistors, the THD for different high-pass corner frequencies were tested. It is seen that the maximum THD occurs when the high-pass corner frequency is set at 3.6 Hz, where the resistance of the tunable pseudo-resistor reaches its minimum. The THD at this worst case is less than 0.6% even when the PGA approaches rail-to-rail output swing. The excellent THD performance is a result of the balanced tunable pseudo-resistor that reduces the signal distortion in the feedback path. It is obvious from Fig. 18 that the THD improves consistently when the high-pass cut-off frequency decreases. This is because with the increase of the tunable resistance, the feedback path becomes dominated by the feedback capacitor, which is implemented by passive

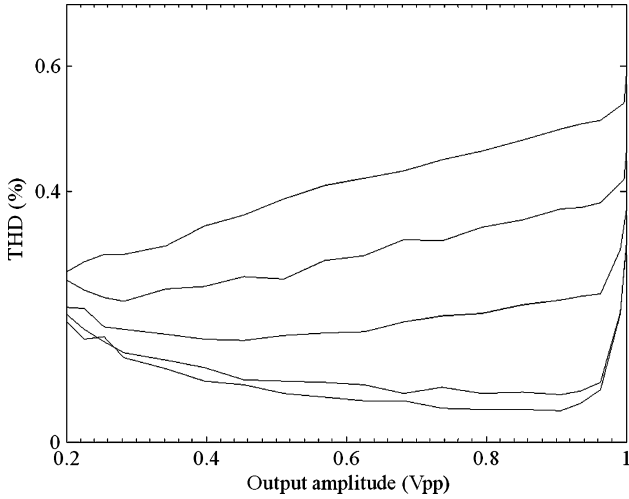


Fig. 18. Total harmonic distortion of the analog front-end versus output amplitude, for different high-pass cutoff frequencies. The cutoff frequency for each curve is (from upper to lower) 3.6 Hz, 2.2 Hz, 1.4 Hz, 0.25 Hz, and 4.5 mHz respectively.

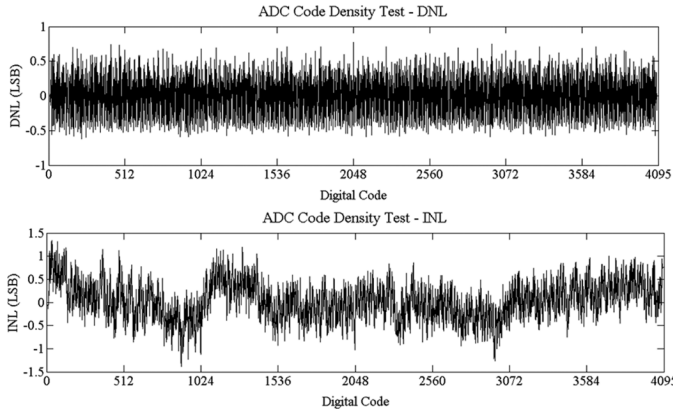


Fig. 19. Code density tests of the SAR-ADC: DNL (upper) and INL (lower).

elements and has better linearity than the nonlinear tunable pseudo-resistor. For large resistance settings such as at 0.25 Hz and 4.5 mHz cut-off frequencies, the harmonic peaks are so small that most of them are buried under the noise floor when the output amplitude is smaller than 0.9 V. This leads to continuously decreasing THD readings in this output range due to the improved SNR (signal-to-noise ratio).

B. ADC Backend

The power consumption of the ADC is measured to be 230 nW. In testing the static nonlinearity performance of the ADC, slow linear voltage ramps were applied to the ADC and the output was captured by an Agilent 1672G logic analyzer. Code density analysis shown in Fig. 19 reveals that the differential nonlinearity (DNL) of the ADC is less than $-0.6/+0.8$ LSB, and the integral nonlinearity (INL) is less than ± 1.4 LSB.

An 11 Hz, $0.95 V_{p-p}$ sinusoidal signal was applied to assess the dynamic performance of the ADC. Fig. 20 depicts the 32,768-point FFT analysis result of the ADC. The second harmonic distortion peak is measured at -74 dB. The calculated

TABLE II
MEASURED PERFORMANCE SUMMARY OF THE SENSOR INTERFACE CHIP

Supply Voltage	1 V
Process Technology	0.35 μ m CMOS
Current (TB-FEA)	33 ~ 337 nA
Midband Gain	45.6 / 49 / 53.5 / 60 dB
High-pass corner frequency	4.5 mHz ~ 3.6 Hz
Low-pass corner frequency	31 ~ 292 Hz
Input Referred Noise	2.5 μ V _{rms} (0.05 Hz ~ 460 Hz)
Noise Efficiency Factor	3.26
THD @ full output swing	0.6%
CMRR	≥ 71.2 dB (Below 300 Hz)
PSRR	≥ 84 dB (Below 300 Hz)
ADC DNL / INL	$< \pm 0.8$ LSB / $< \pm 1.4$ LSB
ADC ENOB	10.2
Total Power Consumption	445 nW ~ 895 nW

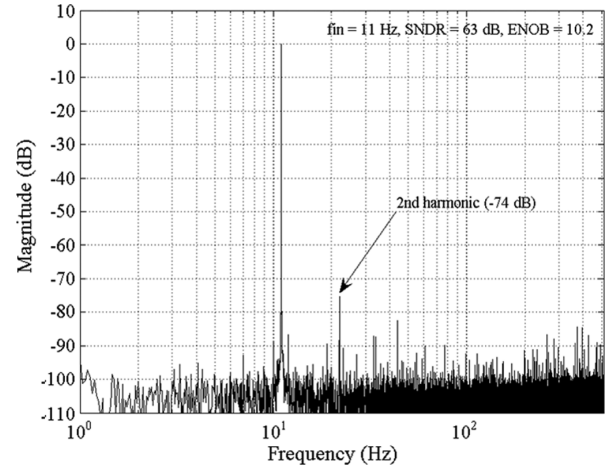


Fig. 20. 32,768-point FFT result with an 11 Hz sinusoidal input signal.

signal to noise plus distortion ratio (SNDR) is 63 dB, corresponding to an ENOB of 10.2 bit.

The relaxation oscillator was measured to have 3 μ s_{rms} jitter running at 30 kHz, and power consumption of 53 nW. Conservatively, three times of this amount of jitter ($\pm 3\sigma$ estimation in a normal distribution) can result in a worst-case tracking error of less than 4.3 mV and an effective resolution of over 7.5 bit, which is sufficient for QRS detections.

C. System Benchmarks

Table II summarizes the measured performance of the sensor interface chip. The proposed design shows improvements in various aspects. The TB-FEA achieves a NEF of 3.26, the lowest among the state-of-the-art designs, indicating an optimum noise-to-power trade-off. To conserve power, different acquisition modes can be selected according to target applications. For example, the chip can operate in two modes in an ECG device. Under the narrowband mode for heart rate detection, where the relaxation oscillator is turned on and the system bandwidth is set to 31 Hz, the overall chip consumes only 445 nA. Under the wideband mode for ECG data recording, the bandwidth is adjusted to 292 Hz and a crystal oscillator is used, which increases the total current consumption to 895 nA.

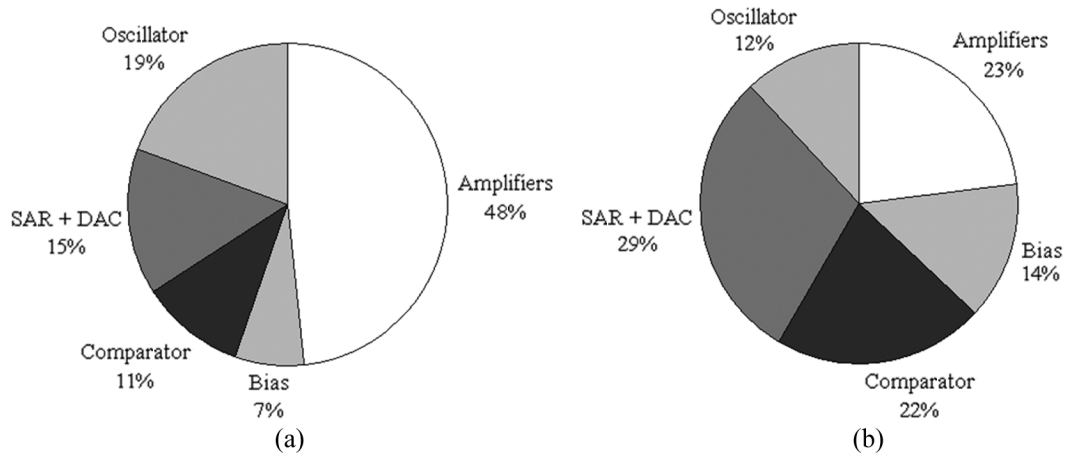


Fig. 21. System power distributions for (a) wideband mode and (b) narrowband mode.

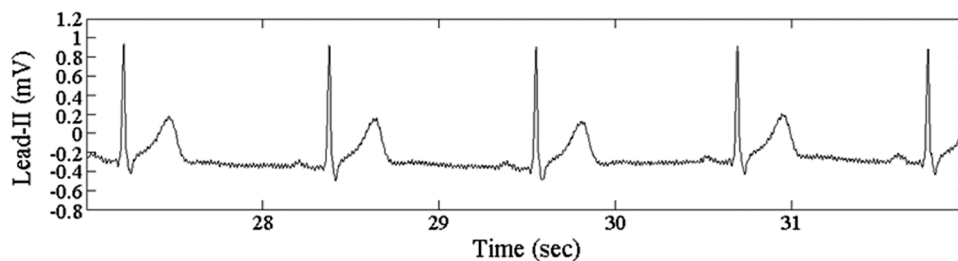


Fig. 22. Recorded human ECG (Lead-II) by the prototype chip.

The system power distributions among these functional blocks for the two modes are shown in Fig. 21.

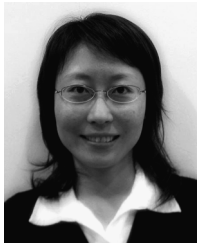
The flexible mode control scheme allows maximum utilization of the tight power budget in a wireless biomedical sensor node. The ultra-low-power consumption of sub- μW for the overall system indicates a fully optimized system structure, which ensures the possibility of portable/implantable use and long term monitoring. To demonstrate the functionality of the chip, a human ECG signal captured using the developed prototype chip is shown in Fig. 22.

VI. CONCLUSION

This paper has demonstrated a sub-microwatt fully integrated programmable biomedical sensor interface chip. A low power system architecture that employs two amplifiers is introduced to optimize the power efficiency. A novel balanced tunable pseudo-resistor effectively reduces the THD to 0.6% at rail-to-rail output swing. A flip-over-capacitor scheme is chosen in the PGA feedback loop to eliminate the low frequency zero-pole pair that appears in the conventional programmable gain control topology. The entire sensor interface chip consumes minimum of 445 nW power at 1 V supply. With features of fully tunable bandwidth and programmable gain, the chip is able to meet requirements of different biomedical applications.

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